

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` system calls and practise the `Strace` command.

```
vida@LAPTOP-4B7UUAUH:~$ nano prac2.c
vida@LAPTOP-4B7UUAUH:~$ gcc prac2.c
vida@LAPTOP-4B7UUAUH:~$ ./a.out
Done, check with 'cat copy.txt' command
vida@LAPTOP-4B7UUAUH:~$ cat copy.txt
helloo
vida@LAPTOP-4B7UUAUH:~$ |
```

- The program uses
- `open()` → opens `sample.txt`
- `read()` → reads data
- `write()` → displays data
- `close()` → closes the file

```

vida@LAPTOP-4B70UUAUH:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffffac0743a8 /* 25 vars */) = 0
brk(NULL) = 0x62f23267e000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x71635b609000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x71635b604000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x71635b5d1000
mmap(0x71635b5d8000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x71635b5d8000
mmap(0x71635b5f9000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x71635b5f9000
mmap(0x71635b600000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x71635b600000
mmap(0x71635b602000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x71635b602000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x71635b5a3000
mmap(0x71635b5a7000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x71635b5a7000
mmap(0x71635b5cb000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x71635b5cb000
mmap(0x71635b5cf000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x71635b5cf000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0

```

